

An explicit weighted digit construction for small sumsets and large difference sets

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Abstract

The small-sumset, large-difference-set problem asks how large $|E - F|$ can be when $|E + F| \leq K|E|$, for fixed $K > 1$ and arbitrarily large finite integer sets E, F . We give a weighted digit construction for its exponent θ . Equal counts of opposite differences give a lower bound for the difference set, while a product-weight estimate bounds the sumset from above. The resulting formula applies to any finite digit alphabet. Four-digit blocks in radix 32, weighted by the sum of their source digits, give $\theta > 1.18565$. The smaller alphabet $\{0, 3, 4, 6, 7, \dots, 16\}$ with geometric weights $(16/19)^a$ gives a simpler proof of $\theta > 1.1855$. Both bounds have exact integer certificates. They exceed Zheng's published lower bound of 1.173077. The proof uses the finite-set transfer inequality of Gyarmati, Hennecart and Ruzsa and does not require optimality of the chosen parameters.

Keywords. sumset; difference set; weighted digits; entropy; exact certificate. **MSC 2020.** 11B75; 05A16.

1 The exponent and the finite-set reduction

Let θ be the supremum of the real numbers t with the following property: for every fixed $K > 1$, there is a constant $c(K) > 0$ and arbitrarily large finite sets $E, F \subset \mathbb{Z}$ satisfying

$$|E + F| \leq K|E|, \quad |E - F| \geq c(K)|E + F|^t. \quad (1)$$

Here $E \pm F = \{e \pm f : e \in E, f \in F\}$, and all logarithms are natural. The constant may depend on the fixed exponent t as well as on K .

The lemma in Section 2 of Gyarmati, Hennecart and Ruzsa [1] gives a finite-set transfer inequality: for every finite $U \subset \mathbb{Z}_{\geq 0}$ with $0 \in U$, $\max U > 0$ and $|U - U| < 2 \max U + 1$,

$$\theta \geq 1 + \frac{\log(|U - U|/|U + U|)}{\log(2 \max U + 1)}. \quad (2)$$

For each such U , its exponent is attained for every fixed $K > 1$ and arbitrarily large sets. The strict difference-count hypothesis matters; we will ensure it by dilating each encoded set. A family whose difference-to-sum ratio grows exponentially relative to its diameter then gives a lower bound for the supremum θ .

Zheng uses integer vectors whose coordinates lie in a full interval and whose coordinate sum is bounded. His large-deviation estimates give the lower bound 1.173077 [2]. We allow a smaller coordinate alphabet and general nonnegative digit costs. Matching the counts of a difference

*Developed with AI research agents. The proof is accompanied by a separate verifier using exact integer arithmetic.

and its negative makes the costs of its two representatives exactly equal. This symmetry gives a direct counting argument without optimizing a large-deviation rate function. The constructions below improve the bound in Zheng's stated theorem.

A separate problem concerns the one-set quantity

$$C(A) = \frac{\log(|A + A|/|A|)}{\log(|A - A|/|A|)}, \quad A \subset \mathbb{Z}, \quad |A| > 1,$$

whose supremum is 2 by Lin and Li [3]. Numerical comparisons must distinguish $C(A)$ from θ . The ‘‘Sets’’ comparison in Table 1 of Beck, Ogren and Kobren [4] reports a value of 1.21 from Hyra, while its Appendix A, equation (7), and Appendix D use the objective considered here. Lin and Li [3, Section 3] identify that 1.21 search value with $C(A)$. The comparison therefore does not establish a 1.21 lower bound for θ .

2 A weighted digit construction

Fix an integer $B \geq 1$ and a digit alphabet $\mathcal{A} \subseteq \{0, \dots, B\}$ containing at least two elements. Attach a positive weight w_a to each $a \in \mathcal{A}$. Multiplying every weight by a common factor changes neither of the ratios below, so assume $\max_{a \in \mathcal{A}} w_a = 1$. Put

$$\mathcal{S} = \mathcal{A} + \mathcal{A}, \quad S_s = \max_{a+b=s} w_a w_b, \quad P = \sum_{s \in \mathcal{S}} S_s, \quad (3)$$

$$\mathcal{D} = \mathcal{A} - \mathcal{A}, \quad D_d = \max_{a-b=d} w_a w_b, \quad Q = \sum_{d \in \mathcal{D}} D_d. \quad (4)$$

All pairs in these maxima belong to $\mathcal{A}^2$; only represented sums and differences are included.

These quantities use the established max-convolution $(f \star g)(z) = \max_{x+y=z} f(x)g(y)$ of finitely supported nonnegative functions; see Matolcsi, Ruzsa, Shakan and Zhelezov [5, Definition 8.1]. Extending w by zero outside $\mathcal{A}$ and putting $\tilde{w}(x) = w(-x)$ gives $P = \|w \star w\|_1$ and $Q = \|w \star \tilde{w}\|_1$. Their Theorem 8.6 relates a weighted max-convolution parameter to an induced tripling parameter. The argument below estimates the reflected quotient Q/P and retains the encoding diameter needed in (2).

Theorem 1. *If $Q > P$, then*

$$\theta \geq 1 + \frac{\log(Q/P)}{\log(2B+1)}. \quad (5)$$

Proof. Write $c_a = -\log w_a \geq 0$ and $\tau_d = -\log D_d$. Choose a pair (a_d, b_d) attaining D_d for each positive $d \in \mathcal{D}$, and set $(a_{-d}, b_{-d}) = (b_d, a_d)$. For $d = 0$, use (a_*, a_*) , where $w_{a_*} = 1$. Then $\tau_0 = 0$ and $\tau_{-d} = \tau_d$. Define

$$p_d = \frac{D_d}{Q}, \quad r = \frac{1}{2} \sum_{d \in \mathcal{D}} p_d \tau_d, \quad V_m = \left\{ x \in \mathcal{A}^m : \sum_{i=1}^m c_{x_i} \leq rm \right\}.$$

The probabilities p_d are positive and satisfy $p_{-d} = p_d$.

Difference count. For $d \neq 0$, take $n_d = \lfloor mp_d \rfloor$ and put $n_0 = m - \sum_{d \neq 0} n_d$. Hence $n_d = n_{-d}$ and $n_d/m \rightarrow p_d$ for every d . Consider any ordering $z \in \mathcal{D}^m$ of this multiset of differences. Replace each coordinate d by the selected pair (a_d, b_d) , obtaining vectors x, y with $x - y = z$. Their two costs are equal, since opposite differences occur equally often. Each cost is

$$\sum_{\substack{d \in \mathcal{D} \\ d > 0}} n_d (c_{a_d} + c_{b_d}) \leq m \sum_{\substack{d \in \mathcal{D} \\ d > 0}} p_d \tau_d = mr.$$

The zero coordinates contribute no cost. Thus every such z belongs to $V_m - V_m$. The number of distinct orderings is $m! / \prod_{d \in \mathcal{D}} n_d!$. Stirling's formula gives

$$\log |V_m - V_m| \geq mH(p) - o(m), \quad (6)$$

$$H(p) = - \sum_{d \in \mathcal{D}} p_d \log p_d = \log Q + \sum_{d \in \mathcal{D}} p_d \tau_d = \log Q + 2r. \quad (7)$$

Sum count. If $z = x + y$ for $x, y \in V_m$, the defining maximum in (3) implies

$$\prod_{i=1}^m S_{z_i} \geq \prod_{i=1}^m w_{x_i} w_{y_i} \geq e^{-2rm}.$$

Summing these nonnegative products over all of $\mathcal{S}^m$ yields P^m . Consequently

$$|V_m + V_m| \leq e^{2rm} P^m. \quad (8)$$

Combining (6)–(8),

$$\log \frac{|V_m - V_m|}{|V_m + V_m|} \geq m \log(Q/P) - o(m). \quad (9)$$

Encoding as integers. Set $L = 2B + 1$ and $\phi_m(x) = \sum_{i=1}^m x_i L^{i-1}$. This map is injective on both $\{0, \dots, 2B\}^m$ and $\{-B, \dots, B\}^m$. Indeed, if two different strings first differ at their highest index j , that term has absolute value at least L^{j-1} , whereas all lower terms together have absolute value at most $2B \sum_{i < j} L^{i-1} = L^{j-1} - 1$.

Let $U_m = \phi_m(V_m) - \min \phi_m(V_m)$. It contains zero, its sum and difference cardinalities equal those of V_m , and

$$2 \max U_m + 1 \leq 2B \sum_{i=1}^m L^{i-1} + 1 = L^m. \quad (10)$$

For all sufficiently large m , the logarithm in (9) is positive, so $M_m = \max U_m > 0$. Set $W_m = \{2u : u \in U_m\}$. This dilation preserves both sum and difference cardinalities and ensures the strict hypothesis of (2), because

$$|W_m - W_m| = |U_m - U_m| \leq 2M_m + 1 < 4M_m + 1 = 2 \max W_m + 1.$$

Moreover, (10) gives $2 \max W_m + 1 \leq 2L^m - 1$. Applying (2) to W_m yields

$$\theta \geq 1 + \frac{m \log(Q/P) - o(m)}{m \log L + \log 2}.$$

Taking $m \rightarrow \infty$ proves (5) for the supremum θ . In particular, every exponent strictly below its right-hand side is attainable in (1); we do not assert attainment of the limit. $\square$

3 Sparse geometric weights

For an alphabet containing zero and geometric weights $w_a = q^a$, with $0 < q < 1$, the two totals have a particularly simple form:

$$P(q) = \sum_{s \in \mathcal{A} + \mathcal{A}} q^s, \quad Q(q) = \sum_{d \in \mathcal{A} - \mathcal{A}} q^{e(d)}, \quad e(d) = \min_{a-b=d} (a+b). \quad (11)$$

Thus both totals are polynomials with nonnegative integer coefficients. The formula follows directly from $w_a w_b = q^{a+b}$ and the strict decrease of q^j with j .

The finite integer family is also explicit. Define

$$\rho(q) = \frac{qQ'(q)}{2Q(q)}, \quad V_m = \left\{ x \in \mathcal{A}^m : \sum_{i=1}^m x_i \leq \lfloor m\rho(q) \rfloor \right\}. \quad (12)$$

The radius r in Theorem 1 is $(-\log q)\rho(q)$, so (12) gives precisely its sets V_m . Multiply their base- $(2B+1)$ encodings by two, as in that proof, to obtain the sets used in the transfer inequality. Zero already belongs to this integer family, and $\rho(q)$ is rational whenever q is rational.

3.1 A one-digit omission

Take $B \geq 3$ and $\mathcal{A}_B = \{0, 2, 3, \dots, B\}$. Its sumset is $\{0, 2, 3, \dots, 2B\}$ and its difference set is $\{-B, \dots, B\}$. For $|d| \geq 2$, the pair $(|d|, 0)$ or its reversal minimizes $a + b$, so $e(d) = |d|$. The difference 1 first occurs as $3 - 2$, giving $e(1) = e(-1) = 5$, and $e(0) = 0$. Therefore

$$P_B(q) = \sum_{j=0}^{2B} q^j - q, \quad Q_B(q) = 1 + 2q^5 + 2 \sum_{j=2}^B q^j. \quad (13)$$

The omitted digit removes one sum, while every difference remains represented. This changes the weighted ratio used by Theorem 1.

Corollary 2. *The exponent in (1) satisfies $\theta > 1.1813$.*

Proof. Set $B = 11$ and $q = 377/500$ in (13). Define the positive integers

$$N_S = \sum_{\substack{0 \leq j \leq 22 \\ j \neq 1}} 377^j 500^{22-j}, \quad (14)$$

$$N_D = 500^{22} + 2 \cdot 377^5 500^{17} + 2 \sum_{j=2}^{11} 377^j 500^{22-j}. \quad (15)$$

Then $Q_{11}(q)/P_{11}(q) = N_D/N_S$. Exact integer exponentiation verifies

$$N_D^{10000} > 23^{1813} N_S^{10000}. \quad (16)$$

The accompanying verifier checks this inequality with integer arithmetic. Taking logarithms in (16) and applying Theorem 1 gives

$$\theta \geq 1 + \frac{\log(N_D/N_S)}{\log 23} > 1 + \frac{1813}{10000} = 1.1813.$$

□

3.2 Omitting the digits 1, 2 and 5

Let $B \geq 8$ and $\mathcal{C}_B = \{0, 3, 4, 6, 7, \dots, B\}$. Its sumset is $\{0, \dots, 2B\} \setminus \{1, 2, 5\}$ and its difference set is $\{-B, \dots, B\}$. The three missing positive digits still occur as differences: $1 = 4 - 3$, $2 = 6 - 4$ and $5 = 8 - 3$. These pairs minimize the sum of their endpoints, giving $e(\pm 1) = 7$, $e(\pm 2) = 10$ and $e(\pm 5) = 11$. Every other nonzero difference uses a digit and zero. Thus

$$P(q) = 1 + q^3 + q^4 + \sum_{j=6}^{2B} q^j, \quad (17)$$

$$Q(q) = 1 + 2 \left(q^3 + q^4 + \sum_{j=6}^B q^j + q^7 + q^{10} + q^{11} \right). \quad (18)$$

Corollary 3. *The exponent in (1) satisfies $\theta > 1.1855$.*

Proof. Take $B = 16$, $q = 16/19$ and $T_j = 16^j 19^{32-j}$. By (17)–(18), $Q(q)/P(q) = M_D/M_S$, where

$$M_S = T_0 + T_3 + T_4 + \sum_{j=6}^{32} T_j, \quad (19)$$

$$M_D = T_0 + 2 \left(T_3 + T_4 + \sum_{j=6}^{16} T_j + T_7 + T_{10} + T_{11} \right). \quad (20)$$

The exact integer check below verifies

$$M_D^{2000} > 33^{371} M_S^{2000}. \quad (21)$$

Consequently Theorem 1 gives

$$\theta \geq 1 + \frac{\log(M_D/M_S)}{\log 33} > 1 + \frac{371}{2000} = 1.1855.$$

□

4 Four-digit blocks with carries

Keep $\mathcal{A} = \{0, 3, 4, 6, 7, \dots, 16\}$ and $q = 16/19$. For $0 \leq k \leq 4$, define the block alphabet and its cost by

$$\mathcal{C}_k = \left\{ \sum_{j=0}^{k-1} a_j 32^j : a_j \in \mathcal{A} \right\}, \quad \lambda \left(\sum_{j=0}^{k-1} a_j 32^j \right) = \sum_{j=0}^{k-1} a_j. \quad (22)$$

For $k = 0$, set $\mathcal{C}_0 = \{0\}$ and $\lambda(0) = 0$. Since every source digit is smaller than 32, each block has a unique representation of length k and its cost is well-defined. Give a block $c \in \mathcal{C}_4$ the weight $w_c = q^{\lambda(c)}$. For example, $96 = 3 \cdot 32$ has cost 3 and weight q^3 . The maximum block value and the outer radix in Theorem 1 are

$$M = 16(1 + 32 + 32^2 + 32^3) = 541200, \quad L = 2M + 1 = 1082401. \quad (23)$$

The weights are positive, at most one, and equal to one at zero.

Sums and differences within a block can have several digit representations. For example, a scalar sum of 32 carries into the next radix-32 position. Each represented integer contributes its largest endpoint-product weight once. We compute that weight by its minimum endpoint cost.

Let $T_k(z)$ and $U_k(z)$ be the minimum of $\lambda(x) + \lambda(y)$ over $x, y \in \mathcal{C}_k$ with $x + y = z$ and $x - y = z$, respectively. Use $+\infty$ when the integer has no representation. Thus $T_0(0) = U_0(0) = 0$, with all other initial values infinite. Put

$$\mathcal{S} = \mathcal{A} + \mathcal{A} = \{0, 3, 4, 6, \dots, 32\}, \quad \mathcal{D} = \mathcal{A} - \mathcal{A} = \{-16, \dots, 16\},$$

and define the scalar difference cost

$$\kappa(d) = \min_{a-b=d} (a+b) = \begin{cases} 7, & |d| = 1, \\ 10, & |d| = 2, \\ 11, & |d| = 5, \\ |d|, & \text{otherwise.} \end{cases}$$

For $0 \leq k < 4$, the exact recurrences are

$$T_{k+1}(z) = \min_{s \in \mathcal{S}} (T_k(z - 32^k s) + s), \quad (24)$$

$$U_{k+1}(z) = \min_{d \in \mathcal{D}} (U_k(z - 32^k d) + \kappa(d)). \quad (25)$$

Indeed, every endpoint pair splits into a pair of shorter blocks and a pair of last digits, so its cost is at least the corresponding expression. Conversely, a minimizing shorter pair and a minimizing scalar pair can be chosen independently and joined to attain that expression. Taking the minimum over all last digits includes every carry and every repeated output.

Write

$$h_S(e) = |\{z : T_4(z) = e\}|, \quad h_D(e) = |\{z : U_4(z) = e\}| \quad (0 \leq e \leq 128).$$

Since $0 < q < 1$, minimizing cost maximizes weight. The quantities in (3)–(4) for the alphabet $\mathcal{C}_4$ are therefore

$$P_4(q) = \sum_{e=0}^{128} h_S(e) q^e, \quad Q_4(q) = \sum_{e=0}^{128} h_D(e) q^e. \quad (26)$$

Define positive integers

$$R_S = \sum_{e=0}^{128} h_S(e) 16^e 19^{128-e}, \quad R_D = \sum_{e=0}^{128} h_D(e) 16^e 19^{128-e}. \quad (27)$$

Both polynomials use the common denominator 19^{128} at $q = 16/19$, so $Q_4(q)/P_4(q) = R_D/R_S$.

Corollary 4. *The exponent in (1) satisfies $\theta > 1.18565$.*

Proof. The accompanying certificate records the two cost histograms and the integers in (27). Its verifier reconstructs both recurrences, checks every histogram entry and integer total, and verifies

$$R_D^{20000} > 1082401^{3713} R_S^{20000}. \quad (28)$$

This inequality implies $Q_4(q) > P_4(q)$. Applying Theorem 1 with alphabet $\mathcal{C}_4$, weights $q^{\lambda(c)}$ and diameter parameter M gives

$$\theta \geq 1 + \frac{\log(R_D/R_S)}{\log 1082401} > 1 + \frac{3713}{20000} = 1.18565.$$

□

The integer family is explicit here as well. Set $\rho_4(q) = qQ'_4(q)/(2Q_4(q))$ and take

$$V_n^{(4)} = \left\{ x \in \mathcal{C}_4^n : \sum_{i=1}^n \lambda(x_i) \leq \lfloor n\rho_4(q) \rfloor \right\}.$$

Encode these vectors in outer radix 1082401 and dilate by two, as in the theorem. The inner radix 32 defines each weighted block; the outer radix preserves all vector sums and differences in the transfer argument.

5 Exact reproduction and scope

The simpler $\theta > 1.1855$ certificate can be checked with the following complete Python program. It uses integer arithmetic and the standard library only.

```
from math import gcd

p, q, B = 16, 19, 16
term = lambda j: p**j * q**(2*B-j)
S = term(0) + term(3) + term(4) + sum(term(j) for j in range(6, 33))
D = term(0) + 2*(term(3) + term(4) + sum(term(j) for j in range(6, 17))
    + term(7) + term(10) + term(11))
g = gcd(S, D)
if (D//g)**2000 <= 33**371 * (S//g)**2000:
    raise ArithmeticError("The certificate failed")
print("Verified: theta > 1.1855")
```

The accompanying `verify.py` also constructs the maxima in (3)–(4) directly from all ordered digit pairs and compares them with the displayed polynomial formulas. The integer inequality, rather than a rounded logarithm or a numerical optimizer, certifies the stated decimal threshold.

For the four-digit bound, `carry-certificate.json` contains the two 129-entry histograms, their integer totals and the fixed parameters. From the accompanying source directory, run

```
python3 -B carry_verify.py
python3 -B -m unittest -v test_carry_verify
```

The verifier implements (24) and (25) in integer cost arrays and checks (28). The controls construct small block alphabets by their endpoint tuples and compare every sum and difference cost with an exhaustive endpoint-pair calculation. They also reject a corrupted histogram and exercise strict inequality, equality and invalid inputs. All reproduction programs use Python 3.10 or later and the standard library.

The theorem holds for every fixed alphabet and every set of positive weights satisfying $Q > P$. The selected parameters need not maximize its right-hand side. Nor does a finite search over alphabets determine the optimal exponent in (1). This note records the construction, its proof and an exact certificate. It does not assert that the bound or method is new, or that the parameters attain a current record.

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